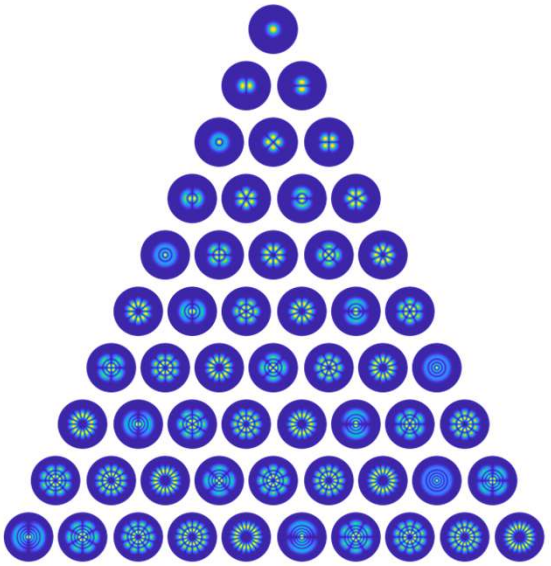
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 10



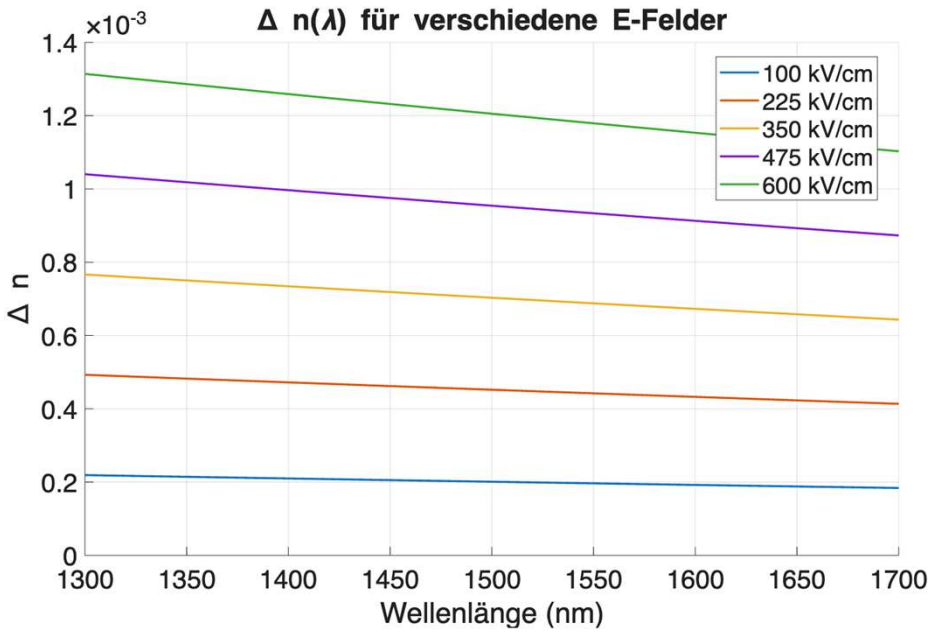
Prof. Dr.-Ing. Georg Rademacher

Modulators

University of Stuttgart
Germany



Phase modulation: Pockels-effect



Phase modulation: Pockels-effect

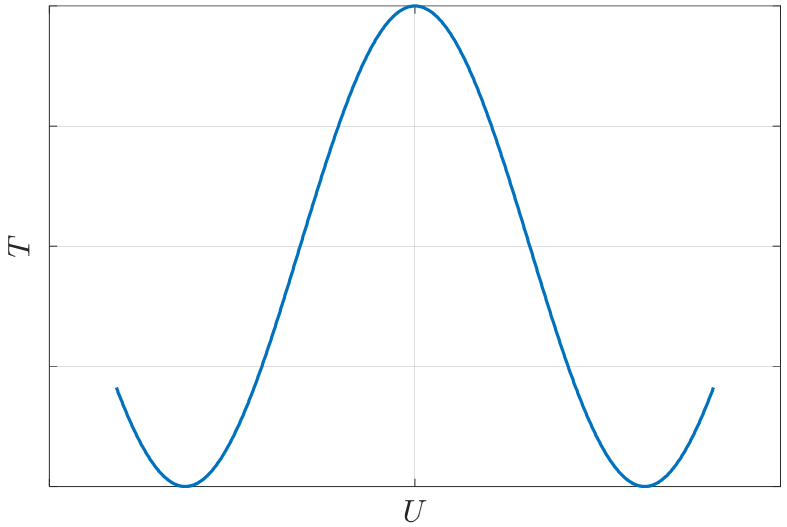
University of Stuttgart
Germany



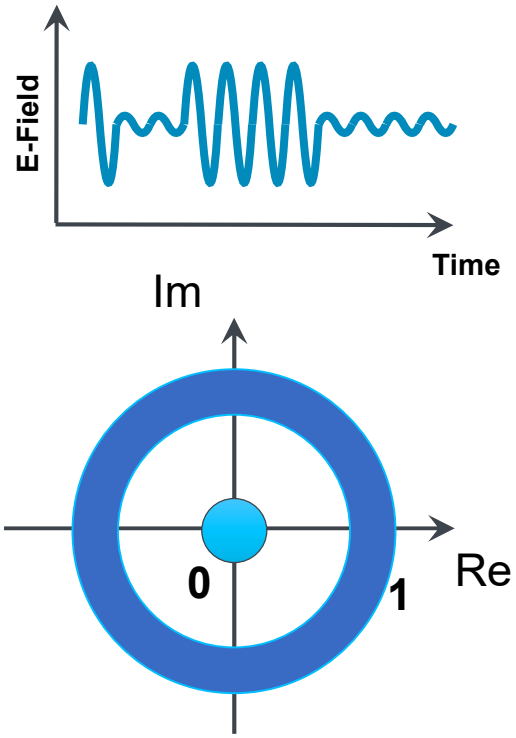
Mach-Zehnder Interferometer

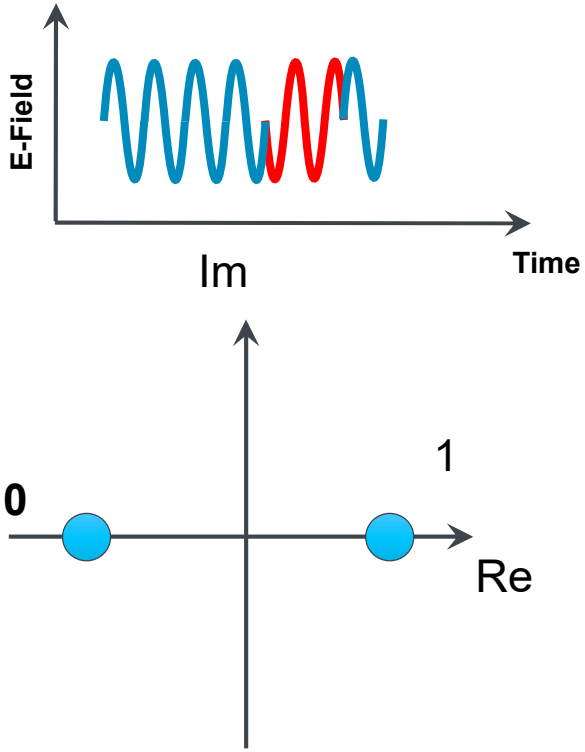


Modulators: Mach Zehnder Interferometer (MZI)

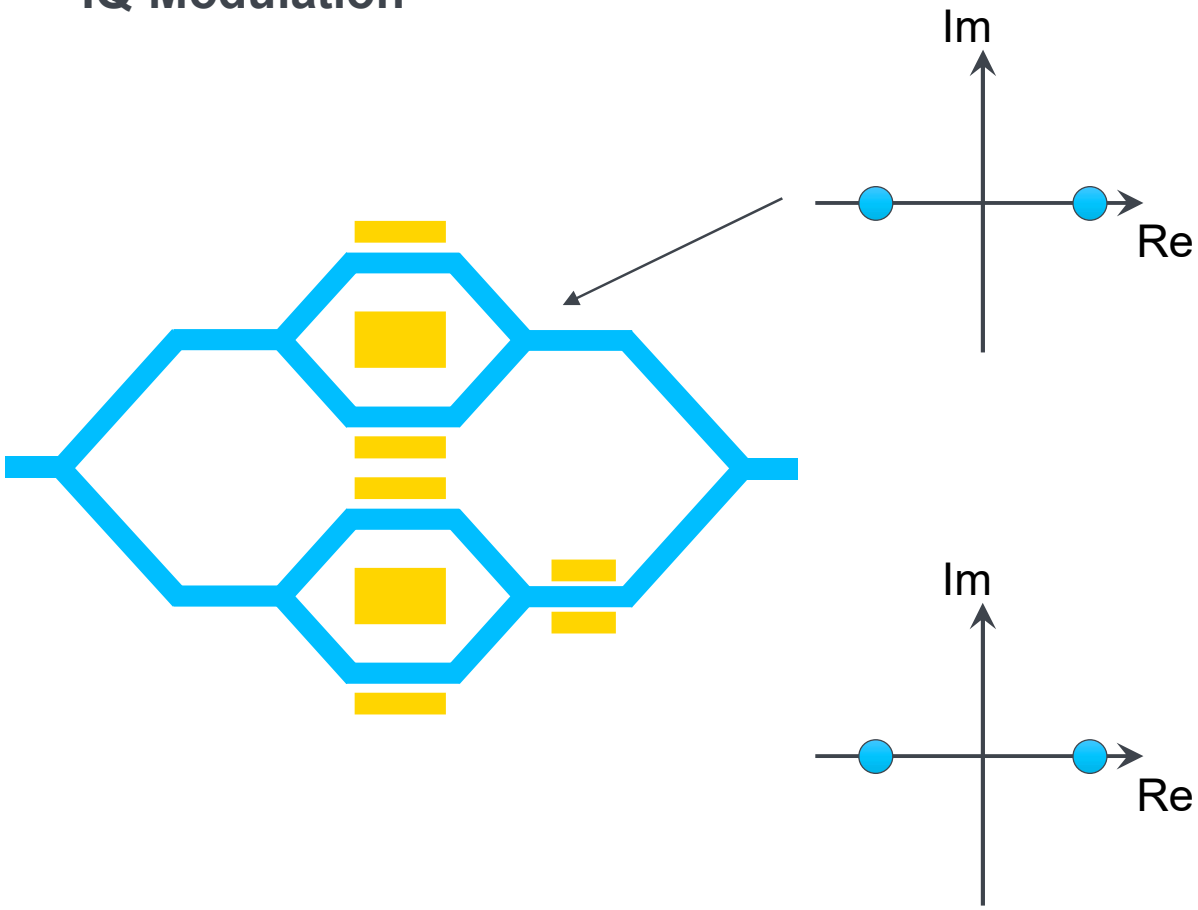


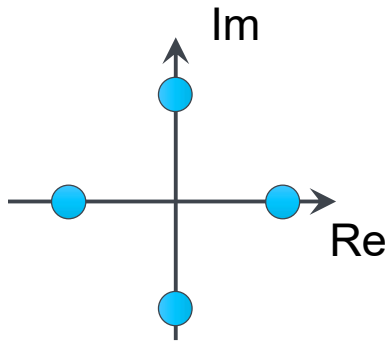
IQ-Modulation





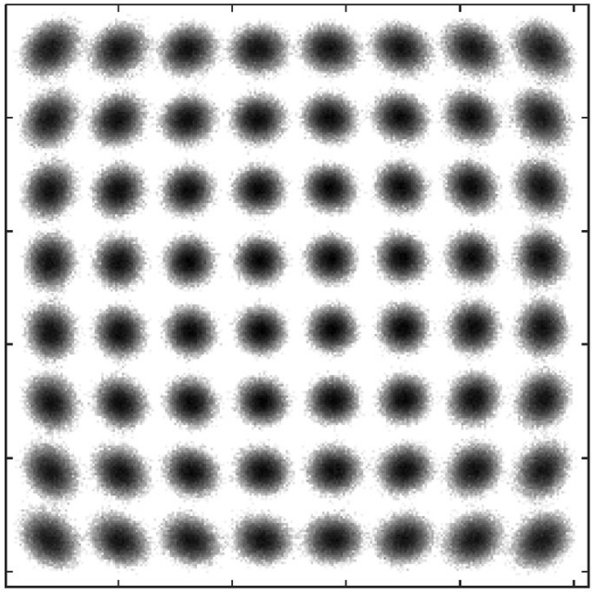
IQ-Modulation

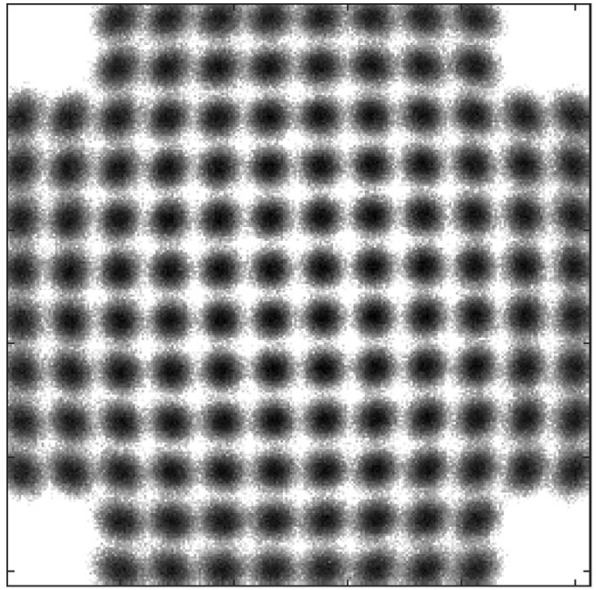


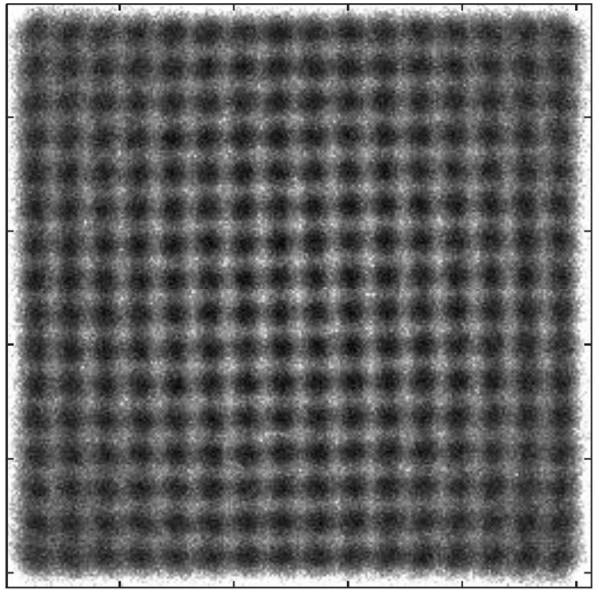


Higher order QAM: Measurements

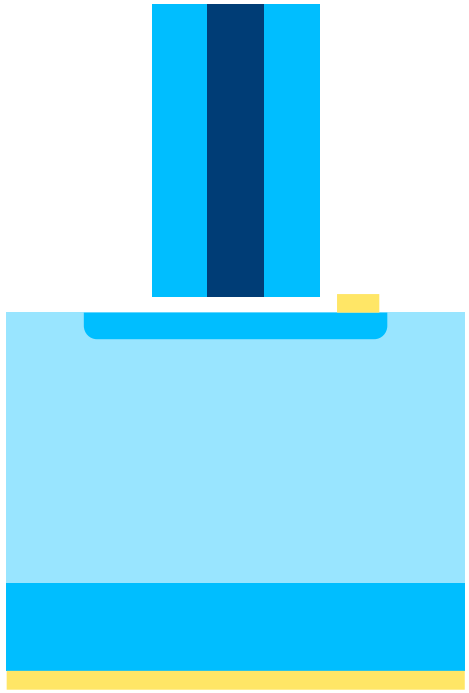
University of Stuttgart
Germany

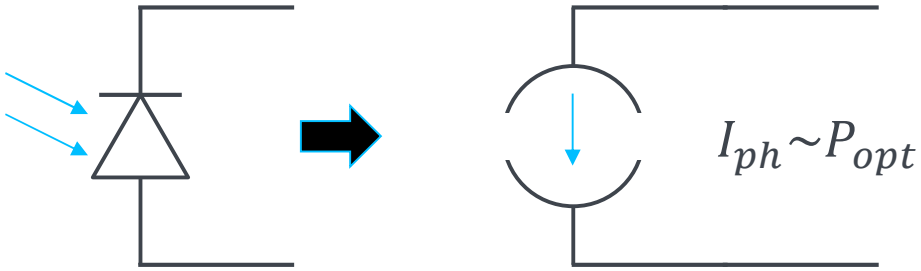




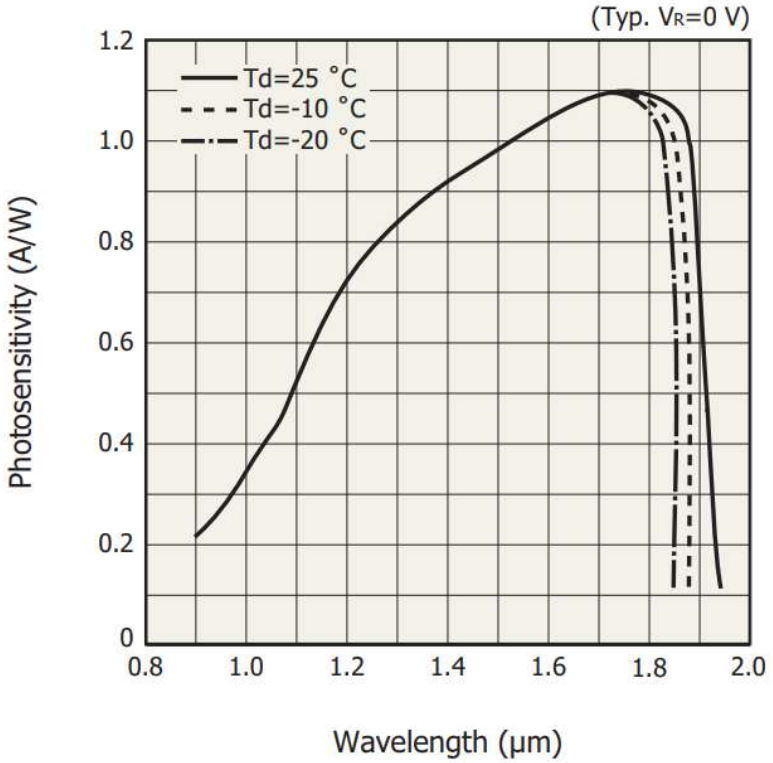


PIN-Photodiodes



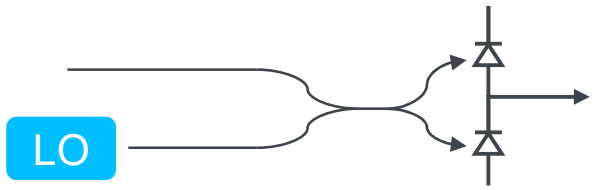


PIN-Photodiodes



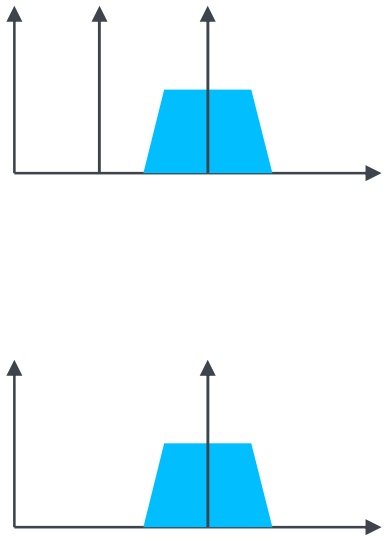
Hamamatsu InGaAs PIN Photodiode

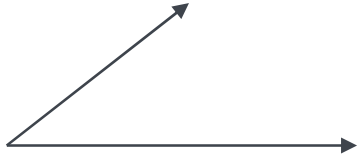
Coherent Detection

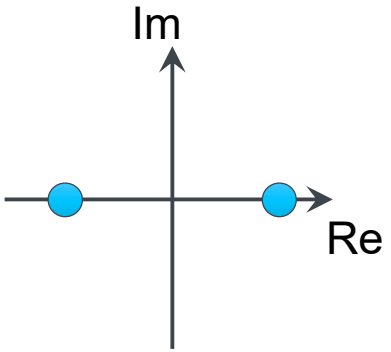


Coherent Detection

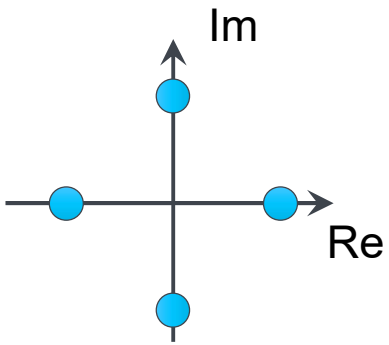
Homodyne Receivers

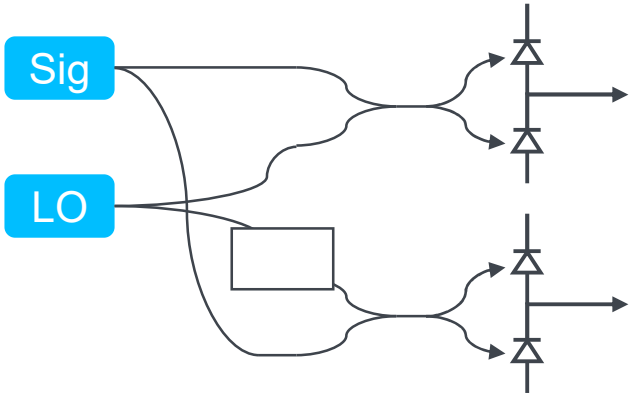






Homodyne Receivers: I and Q





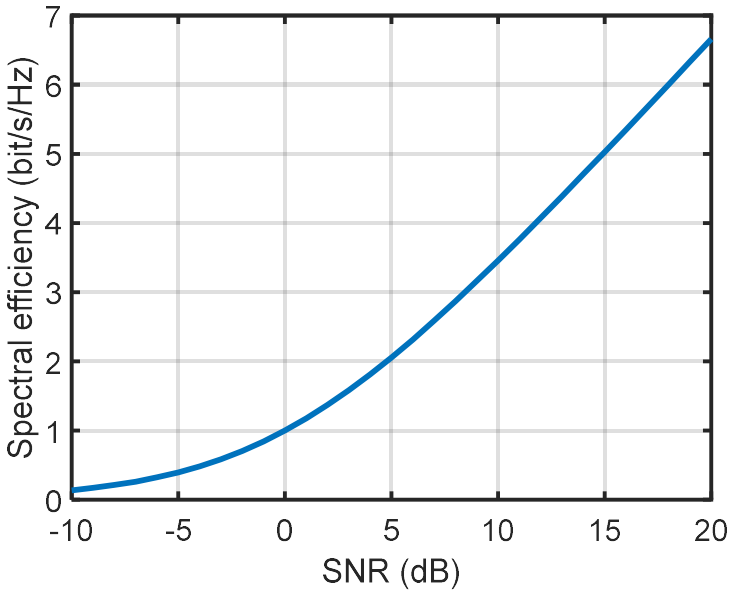
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 34, NO. 1, JANUARY 1, 2016

Fundamentals of Coherent Optical Fiber Communications

Kazuro Kikuchi, *Fellow, IEEE, Fellow, OSA*

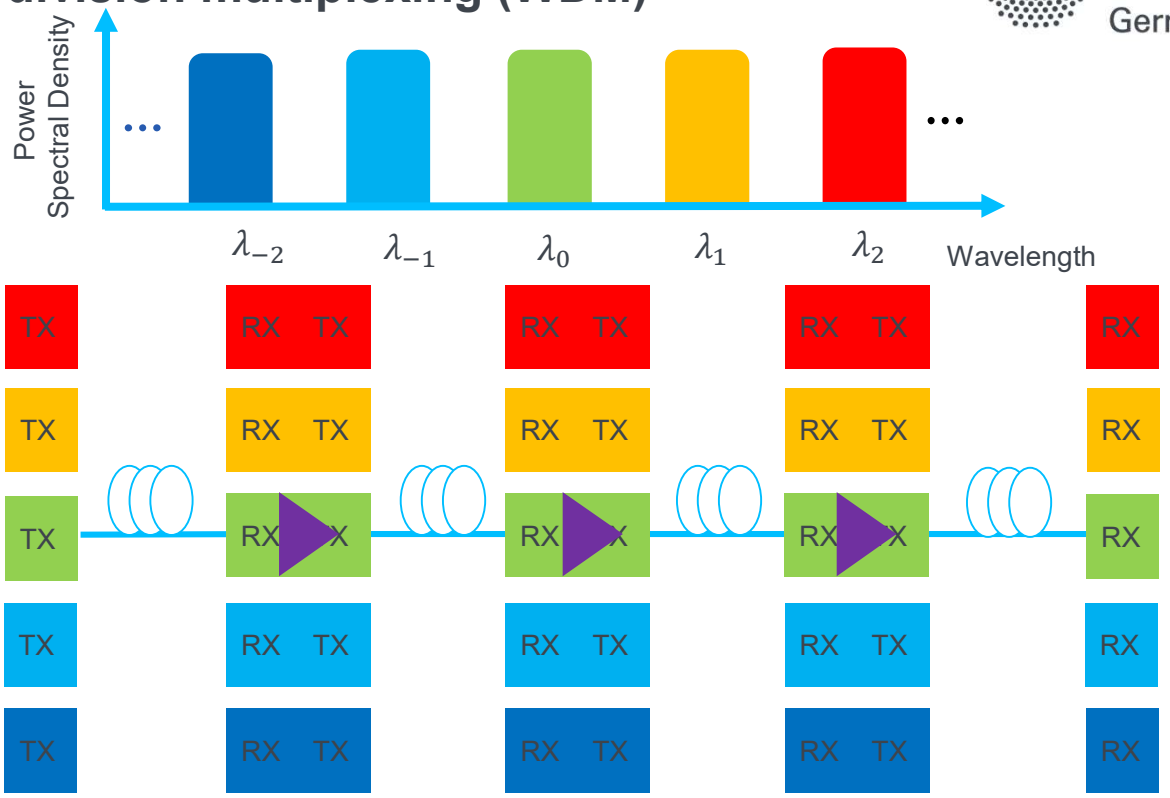
(Tutorial Review)

Optical transmission link



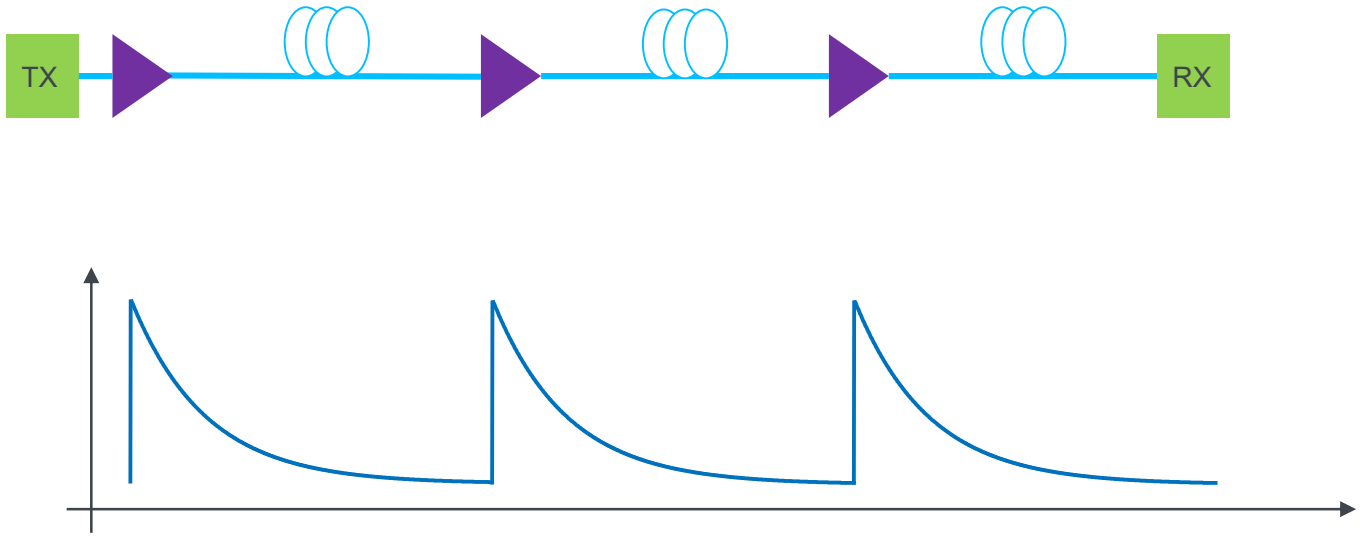
Wavelength division multiplexing (WDM)

University of Stuttgart
Germany



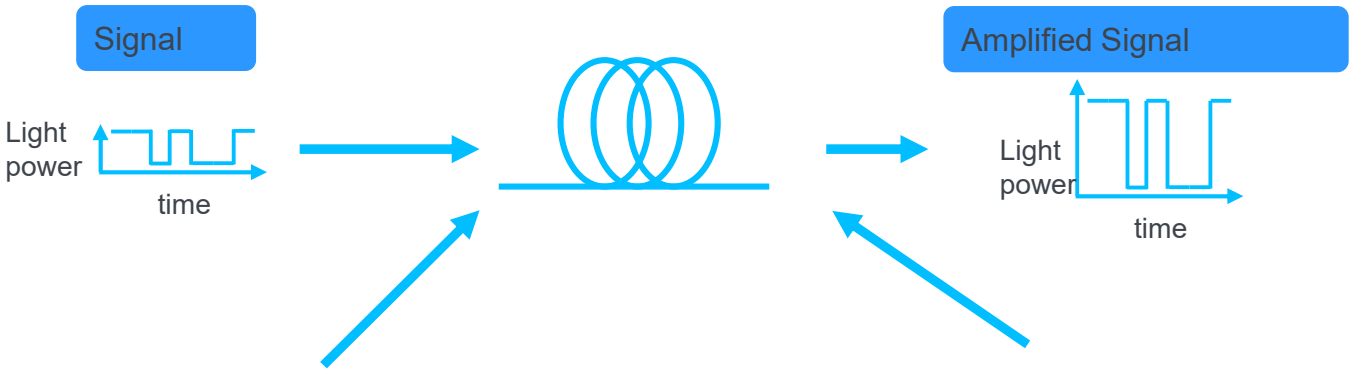
Power profile in amplified transmission link

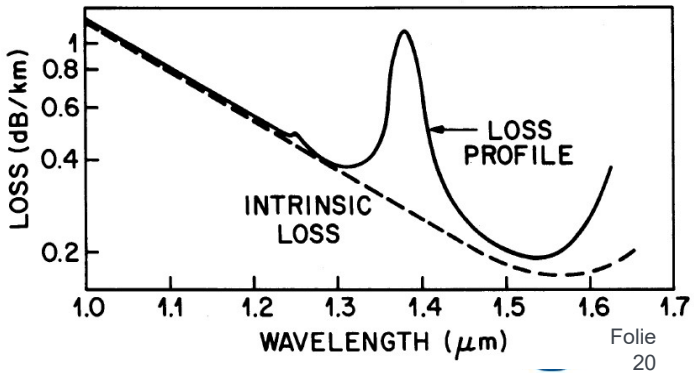
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Germany



Erbium doped fiber amplifier (EDFA)

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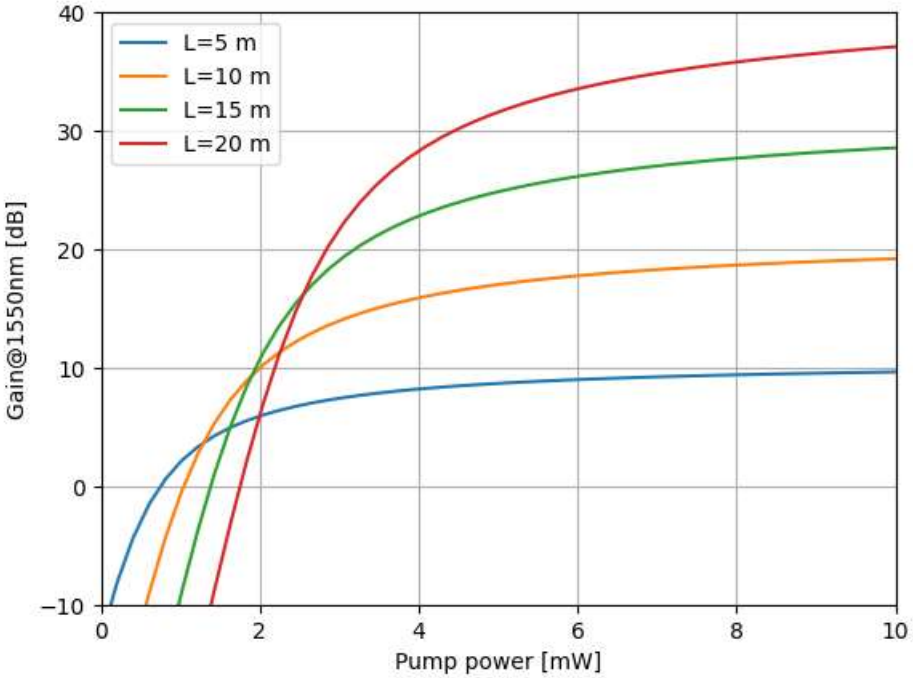


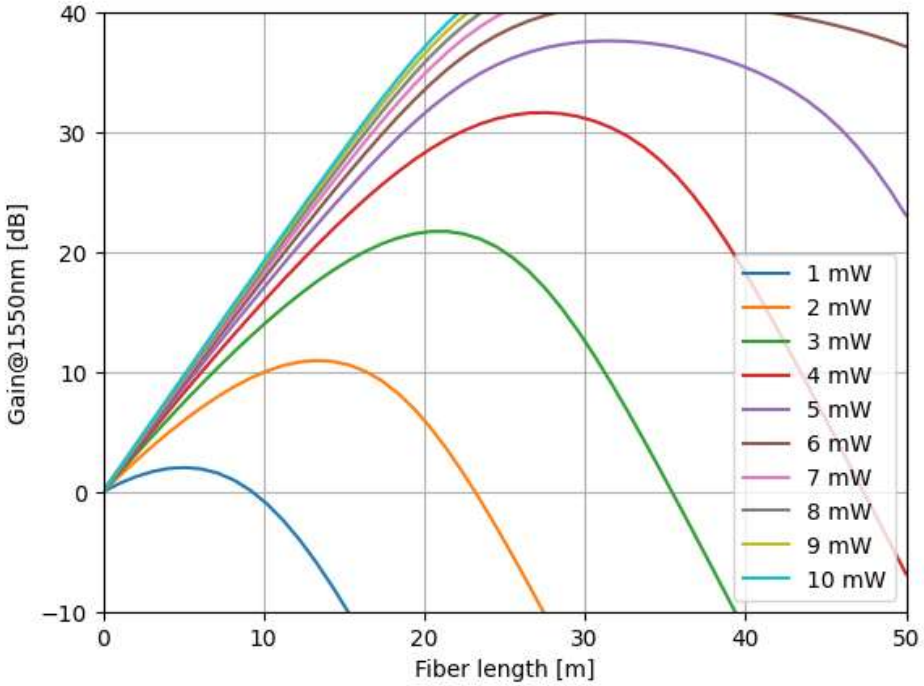
Erbium-doped fiber amplifier (EDFA)



Pump and signal power evolution

Gain vs. pump power and fiber length

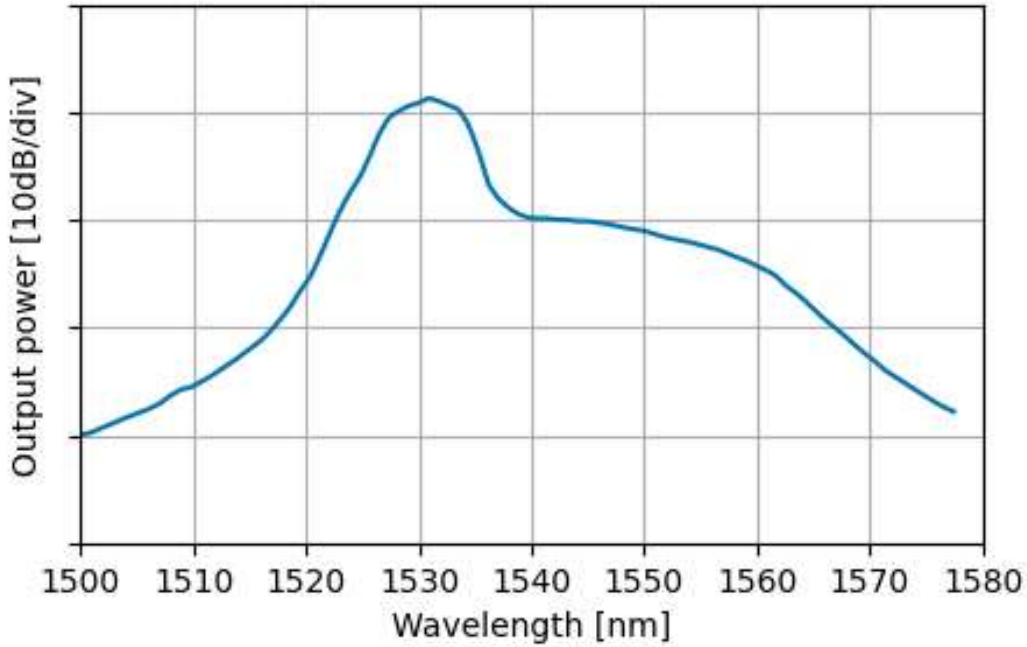




EDFA Noise

EDFA Noise

EDFA Noise


Output power [10dB/div]
1500
1510
1520
1530
1540
1550
1560
1570
1580
Wavelength [nm]

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Institute of electrical and optical communications

Thank you!



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